

3-3 Conservative Forces and Potential Energy



Source / 來源

- 選修物理(二) 課堂講義 (4624R51) 教用版，第3章 功與能量，§ 3-3 保守力與位能，pp. 124–133
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 8: Potential Energy and Conservation of Energy
- Young & Freedman, *University Physics with Modern Physics*, Ch 7: Potential Energy and Energy Conservation

Examples

Example 1 — Gravitational Work on a Chain (*p. 126*)

A uniform chain of length ℓ and mass m hangs with one third of its length over the edge of a table. The chain is slowly pulled completely onto the table. Find the work done by gravity. Take the tabletop as $U_g = 0$ and use g .



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Solution

Try It 1 — Folded Uniform Rope (*p. 126*)

A uniform rope of length ℓ and mass m is suspended from point O . Point C is brought to O so that the rope folds into segments OBC and CA , with $OB = 2\ell/5$ and $CA = \ell/5$. Find the work done by gravity.

 Solution

Example 2 — Elastic Potential Energy (*p. 128, 96 指考*)

An object attached to an ideal spring is pulled from equilibrium until the spring is stretched by $2x$, where its elastic potential energy is U . It is released and passes equilibrium; when the spring is compressed by x , what is its elastic potential energy?

 Solution

Try It 2 — Spring Bent at Its Midpoint (*p. 128*)

A spring has natural length 0.80 m. Stretching it to 1.80 m requires 100 J. Its two ends are then fixed and its midpoint is pulled 0.30 m perpendicular to the original line. Find the total elastic potential energy.

Quick Check

Gravitational Potential Energy

Q1. Ball A has mass 8 kg and uses height 5 m as its zero level. Ball B has mass 4 kg and uses height 10 m as its zero level. If their gravitational potential energies are equal, what is h_A/h_B ?

(A) 4 (B) $1/4$ (C) 2 (D) $1/2$ (E) 6

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Q2. A uniform rod of length ℓ is pivoted at one end. It is slowly rotated from vertical through 60° . How much work must an external force do against gravity?

(A) $mg\ell/8$ (B) $mg\ell/4$ (C) $mg\ell/2$ (D) $3mg\ell/4$ (E) $mg\ell$

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 Solution

Q3. A uniform rod of length L and weight W lies flat on the ground. It is raised upright and placed on a table of height h . What minimum work is required?

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 Solution

Elastic Potential Energy

Q4. A vertical spring stretches by x_0 when it supports a mass m . The same spring is horizontal on a frictionless table. If the mass is pulled $3x_0/2$ from equilibrium and released, what is the spring potential energy when it has moved a distance x toward equilibrium?

(A) $mgx^2/(8x_0)$ (B) $mgx^2/(6x_0)$ (C) $mgx^2/(4x_0)$ (D) $mgx^2/(2x_0)$ (E) $mgx^2/(2x_0)$

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 Solution

Q5. In a CO molecule, the equilibrium separation is 1.13×10^{-10} m. At 1.15×10^{-10} m the attractive force is 3.74×10^{-10} N. Assuming Hooke's law, how much does the molecular potential energy increase?

(A) 2.46×10^{-10} J (B) 4.30×10^{-20} J (C) 7.58×10^{-22} J (D) 5.74×10^{-22} J (E) 3.74×10^{-22} J

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 Solution

Challenge

Gravitational Potential Energy

Q1. A 1 m uniform rope weighs 10 kg; 40 cm hangs over a table edge. What minimum work pulls 20 cm back onto the table? Take $g = 10 \text{ m/s}^2$.

(A) 2 J (B) 4 J (C) 6 J (D) 8 J (E) 12 J

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 Solution

Q2. A mass m falls freely. How much gravitational potential energy is lost during the t th second?

(A) $\frac{1}{2}mgt^2$ (B) $\frac{1}{2}mg(t-1)^2$ (C) $\frac{1}{2}mg(2t-1)$ (D) mgt^2 (E) mgt

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 Solution

Rectangular Blocks

Q3. A rectangular box has side lengths a, b and mass m . It is initially lying with side a vertical and b horizontal. What minimum work tips it onto the other side?

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 Solution

Q4. If the box is initially lying with side b vertical and is raised to the orientation with side a vertical, what minimum work is required?

 Solution

Spring-Mass System

Q5. A spring-mass system starts at equilibrium and moves to $x = 1.00$ m. The applied force has an F - x graph that is a triangle of area 0.50 J, while the spring force does -0.25 J. Which statement is correct?

- (A) The net work is zero.
- (B) The spring does $+0.25$ J.
- (C) The system potential decreases by 0.25 J.
- (D) The object's kinetic energy increases by 0.25 J.
- (E) The object's speed decreases.

 Solution

Q6. Two coaxial springs have the same spring constant k and differ in natural length by ℓ . If a compression makes the shorter spring deform by $\ell/2$, what is the total elastic potential energy?

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 Solution

Exam Archive

Q1 — Barbell Energy from a Velocity Graph (*p. 132, 109 學測*)

A 100 kg barbell is lifted vertically. Its velocity–time graph is shown in the source figure (upward is positive; labeled instants 甲–壬). Which statement is correct?

(A) Kinetic energy increases from $t = 0$ to 甲.

(B) Potential energy decreases from 丁 to 戊.

(C) Potential energy decreases from 戊 to 庚.

(D) 己 is the minimum-potential point.

(E) 丙 is the maximum-potential point.

[Figure: Velocity–time graph with labeled instants 甲–壬.]

 Solution

Q2 — Box Pulled up a Rough Incline (*p. 132, 108 指考*)

A box is pulled up a rough incline at constant speed. The pulling force is parallel to the incline. Which statements are correct? (*Select all that apply.*)

- (A) Gravity does negative work.
- (B) The pulling force does positive work.
- (C) Friction does positive work and the normal force does no work.
- (D) The box's kinetic energy does not increase, so total work is zero.
- (E) The box's gravitational potential increases, so total work is positive.

 Solution

Q3 — Force from a Potential-Energy Curve (*p. 133, 105 指考*)

A particle moves between x_1 and x_2 in a potential $U(x)$ that is a parabola with its minimum at x_0 . Which force–position graph is possible?

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 Solution

Q4 — Elastic Potential over One SHM Period (*p. 133, 103 指考*)

For a mass on an ideal spring in one-dimensional simple harmonic motion, which graph best shows elastic potential energy U_s versus time over one period?

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 Solution

Q5 — Skier and Dissipated Energy (*p. 133, 99 指考*)

A 60 kg skier starts from rest at P , reaches R with speed 18 m/s, and descends a net vertical height of 20 m. Find the ratio of energy dissipated by friction to the gravitational potential-energy decrease. Take $g = 9.8 \text{ m/s}^2$.

(A) 1 (B) 0.8 (C) 0.3 (D) 0.2 (E) 0.1

 Solution

Glossary

English	中文	Symbol / Unit
conservative force	保守力	—
non-conservative force	非保守力	
potential energy	位能	U [J]
gravitational potential energy	重力位能	$U_g = mgy$
elastic potential energy	彈性位能	$U_s = \frac{1}{2}kx^2$
zero level / reference level	零位面	$U = 0$
center of mass	質心	—
path independent	與路徑無關	
closed path	封閉路徑	$W_{\text{cons}} = 0$
work by a conservative force	保守力作功	$W_c = -\Delta U$
Hooke's law	虎克定律	$F = -kx$
simple harmonic motion	簡諧運動	SHM
force–position graph	力–位置圖	F – x